# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 1

**Full Syllabus (Chapters 1–13)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): Ignoring air resistance in a model of a falling ball is acceptable for a stone. Reason (R): For a stone, the effect of air resistance is very small compared with its weight. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** Assertion (A): Milk shows the Tyndall effect but salt solution does not. Reason (R): The particles of milk are larger than the particles of a true solution. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**3.** Assertion (A): The law of constant proportions is true for compounds but not for mixtures. Reason (R): In a compound the elements are chemically combined in a fixed ratio. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**4.** Assertion (A): Most alpha particles pass undeflected through a thin gold foil. Reason (R): The nucleus occupies a very small fraction of the atom's volume. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**5.** Assertion (A): A body can have zero displacement but non-zero distance. Reason (R): Distance is the length of the path and displacement is the net change in position. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**6.** Assertion (A): A cricketer moves his hands backward while catching a fast ball. Reason (R): This increases the time of stopping and hence decreases the force on the hands. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**7.** Assertion (A): No work is done when a person pushes against a rigid wall. Reason (R): The displacement of the wall is zero. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**8.** Assertion (A): Sound cannot travel in a vacuum. Reason (R): Sound waves need a material medium for propagation. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**9.** Assertion (A): Red blood cells swell and burst in distilled water. Reason (R): Water enters the cells by osmosis because they lack a cell wall. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** Assertion (A): Grasses grow again after being cut. Reason (R): Intercalary meristem is present at the base of their leaves. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**11.** Assertion (A): Whale is placed in the class Mammalia. Reason (R): It has mammary glands and breathes with lungs. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**12.** Assertion (A): Plants raised by vegetative propagation are genetically identical to the parent. Reason (R): They are produced by mitosis from a single parent. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**13.** Assertion (A): The polar regions are colder than the equator. Reason (R): Sunlight falls at a slant at the poles and is spread over a larger area. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** The mass of the Earth is about 6 x 10^24 kg and that of the Sun about 2 x 10^30 kg. The Sun is about ___ times heavier. 1

**(a)** 3 x 10^4 **(b)** 3 x 10^6 **(c)** 3 x 10^54 **(d)** 3 x 10^5

**15.** A solution has 30 g of solute in 120 g of solvent. 90 g of the solvent is evaporated. The new percentage by mass of solute is: 1

**(a)** 25 per cent **(b)** 20 per cent **(c)** 75 per cent **(d)** 50 per cent

**16.** A compound of A and B contains 75 per cent A by mass. A second compound of the same elements contains 60 per cent A. The ratio of the masses of A that combine with a fixed mass of B in the two compounds is: 1

**(a)** 2 : 1 **(b)** 4 : 3 **(c)** 3 : 2 **(d)** 3 : 1

**17.** **Assertion (A):** Repeating a measurement several times and taking the mean gives a more reliable value than a single reading. 1
**Reason (R):** Random errors in individual readings tend to cancel out when several readings are averaged.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**18.** **Assertion (A):** A mixture of salt and sand can be separated by dissolving it in water, filtering and evaporating the filtrate. 1
**Reason (R):** Sand is insoluble in water, whereas salt is soluble in water.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**19.** **Assertion (A):** The total mass of the products equals the total mass of the reactants in a chemical reaction carried out in a closed vessel. 1
**Reason (R):** Atoms are neither created nor destroyed in a chemical reaction; they are only rearranged.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**20.** **Assertion (A):** In Rutherford's experiment most alpha particles passed straight through the gold foil. 1

**Reason (R):** Most of the space inside an atom is empty.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) 6 × 2 = 12

*Answer in about 20–30 words each.*

**21.** The sides of a rectangle are measured as 12.0 cm and 8.0 cm with an uncertainty of 0.1 cm each. Find the area and the percentage uncertainty. 2

**22.** A solution has density 1.2 g/mL and contains 24 g of solute in 200 mL. Find the mass by mass percentage. 2

**23.** Find the mass of water formed when 4 g of hydrogen reacts with 40 g of oxygen. Which reactant is left over and how much? 2

**24.** Find the percentage of the heavier isotope for an element of average mass 63.5 u with isotopes 63 u and 65 u. 2

**25.** A car travels the first half of the distance at 30 km/h and the next half at 60 km/h. Find the average speed. 2

**26.** A body of mass 2 kg moving with 10 m/s is acted on by a force of 4 N opposite to its motion. Find the time to stop and the distance. 2

## SECTION — C (Short Answer Type) 6 × 3 = 18

*Answer in about 40–60 words each.*

**27.** A student measures a pencil three times as 14.2 cm, 14.3 cm and 14.1 cm. Find the mean length and the greatest deviation from it. 3

**28.** The solubility of a salt is 25 g per 100 g water at 20 degrees Celsius and 60 g per 100 g water at 60 degrees Celsius. Find the mass of crystals obtained when 320 g of saturated solution at 60 degrees Celsius is cooled to 20 degrees Celsius. 3

**29.** A compound contains 1.6 g of oxygen for 1 g of sulphur (S = 32, O = 16). Find the atoms ratio and the formula. 3

**30.** A metal M forms an oxide $M_2O_3$. Its atom has 13 protons. State its configuration, ion and the formula unit mass if its mass number is 27. 3

**31.** A particle starts from rest with a = 4 m/s squared for 5 s, then moves uniformly for 10 s and then decelerates at 2 m/s squared until it stops. Find the total distance. 3

OR

A car of mass 1500 kg is moving on a level road at 72 km/h. The driver applies the brakes and the car stops in 5 s. Find the force and the distance. Compare the force with the weight. 3

**32.** A block of mass 5 kg slides down a rough incline of height 4 m and reaches the bottom at 6 m/s. Find the work done against friction (g = 10). 3

OR

Two sound waves have frequencies 100 Hz and 400 Hz with equal amplitude. Compare their wavelengths, speeds, loudness and pitch in the same medium. 3

## SECTION — D (Long Answer Type) 6 × 5 = 30

*Answer in about 100–150 words each.*

**33.** **Case study.** Kavya measures the length of a pencil with a ruler marked in millimetres. She takes five readings: 14.2 cm, 14.4 cm, 14.3 cm, 14.1 cm and 14.5 cm. Her teacher says that no single reading may be exactly right, so the readings should be combined. 5

(i) What is the least count of her ruler in cm?

(ii) Find the mean length of the pencil.

(iii) Express the mean length in metres.

(iv) Why is the mean more reliable than the first reading, and what is the largest deviation of any reading from the mean?

**34.** **Case study.** A student in Pune dissolves 20 g of common salt in 180 g of water to prepare a solution for an experiment. 5

(i) Name the solute and the solvent.

(ii) Find the mass of the solution.

(iii) Find the mass percentage of salt in the solution.

(iv) How much more water must be added to this solution to make a 5 per cent (by mass) solution?

**35.** A student burns 2.4 g of magnesium in a crucible and finds the oxide mass to be 4.0 g. She repeats with 6.0 g and gets 10.0 g. Show that the results support the law of constant proportions, find the ratio Mg : O and verify the formula MgO (Mg = 24, O = 16). 5

**36.** The atomic masses of two isotopes of an element are 69 u and 71 u and its average atomic mass is 69.8 u. Find the percentage abundance of each isotope and the number of neutrons in each if Z = 31. Name the element. 5

OR

A train starts from rest at station A and stops at station B 3 km away. It accelerates at 0.5 m/s squared for the first part and retards at 1 m/s squared for the last part with no uniform stretch. Find the maximum speed, the total time and the distances in the two parts. 5

**37.** A body of mass 8 kg starts from rest and is acted on by a net force of 24 N. Find the acceleration, the velocity and the distance after 5 s, and the momentum then. 5

OR

A block of mass 2 kg is released from rest at the top of a smooth curved track of height 3.2 m, slides along a rough horizontal surface of length 4 m and stops. Find the speed at the bottom, the friction force and the work done by friction (g = 10). 5

**38.** A source of frequency 440 Hz sends a sound wave in air (344 m/s) towards a wall 172 m away. Find the wavelength, the time for the echo, the number of waves between the source and the wall, and the number of complete waves in the echo path. 5

OR

Design an experiment to find the concentration of the cell sap of potato tissue: test cubes in 0, 2, 4, 6, 8 per cent sugar solutions, mention what to measure, expected results and the use of the graph (zero change in mass). 5

## END OF QUESTION PAPER Total: 80 Marks